

ALL-DEPTH ENUMERATION AND TERMINAL FIBRES FOR TRANSLATION-GCD EROSION OVER FINITE FIELDS

ANONYMOUS

ABSTRACT. Let $q = p^a$ and iterate $T(f) = \gcd(f(x), f(x+1))$ on monic polynomials over $\mathbb{F}_q$. Translation has order p , so the process reaches the invariant core $Q(f) = T^{p-1}(f)$. We determine, for every $0 \leq t \leq p-1$, an exact all-degree generating function for the states whose stabilization depth is at most t . On each nonfixed orbit of irreducible factors, subtracting the minimum exponent leaves a cyclic nonnegative vector with a zero; its depth is its longest positive run. A finite run automaton supplies the local series, and an Euler product over irreducible translation orbits gives the global census. We also prove a unique graded split into an invariant core and the unit fibre $Q^{-1}(1)$. Its generating function is $(1 - qz^p)/(1 - qz)$, yielding the exact-degree and degree-capped fibre over every invariant target. The translation-fixed irreducible counts used in the product are due to Garefalakis and Reis, and the underlying finite orbit-fold mechanism is treated as prior background; the residual claims are the all-depth product and target-refined fibres. Finite checks over $\mathbb{F}_4, \mathbb{F}_8$, and $\mathbb{F}_9$ are falsification controls only. External release remains on hold.

1. SCOPE AND SETUP

Let $q = p^a$ be a prime power and let $\mathcal{M}_q$ be the set of monic polynomials in $\mathbb{F}_q[x]$, including 1. Write

$$\sigma f(x) = f(x+1), \quad T(f) = \gcd(f, \sigma f), \quad Q = T^{p-1},$$

where every gcd is monic. Since $\sigma^p = 1$, the acting subgroup is the prime-subfield copy of $\mathbb{F}_p$, not the full additive group of $\mathbb{F}_q$ when $a > 1$. We use the stabilization depth

$$\text{depth}(f) = \min\{t \geq 0 : T^t(f) = Q(f)\}.$$

Thus fixed states have depth zero.

The invariant ring for this translation is $\mathbb{F}_q[x^p - x]$. Its monic elements have series

$$(1) \quad I_{q,p}(z) = \sum_{h \in \mathcal{M}_q : \sigma h = h} z^{\deg h} = \frac{1}{1 - qz^p},$$

because $h = g(x^p - x)$ uniquely, with g monic. This characterization and the translation-fixed irreducible classification are established in the finite-field invariant literature, especially Garefalakis [1] and Reis [5]; related group-action and invariant-function frameworks include [3, 4, 6].

We state the precise enumerative input. Let

$$(2) \quad N_d(q) = \frac{1}{d} \sum_{e|d} \mu(e) q^{d/e}$$

be the number of degree- d monic irreducibles. Let b_d count those fixed by σ . Then $b_d = 0$ unless $d = pm$, and, if $m = p^v s$ with $(s, p) = 1$,

$$(3) \quad b_{pm} = \frac{p-1}{pm} \sum_{e|s} \mu(s/e) q^{p^v e}.$$

This is exactly Reis's formula [5, Theorem 2(c)], after changing divisor variables, and is *not* a contribution here. Consequently

$$(4) \quad a_d = \frac{N_d(q) - b_d}{p}$$

is the number of length- p translation orbits of degree- d irreducibles. Equations (2)–(4) are owned input to our Euler product.

The elementary identity

$$(5) \quad T^t(f) = \gcd_{0 \leq j \leq t} \sigma^j f$$

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follows by induction. It implies $T^{p-1}(f)$ is invariant and $\text{depth}(f) \leq p-1$. These facts, the sharp clock, the fixed set, and old finite depth tables receive zero contribution credit.

Remark 1.1 (P110 order-dual firewall). An earlier internal note, P110, studies $J(x) = x \vee \sigma x$ on a cyclically acted lattice and obtains the consecutive orbit join, invariant endpoint, finite clock, and recurrent points. Equation (5) is its order-dual meet mechanism in the divisibility lattice. We claim none of those generic semilattice conclusions. The contribution below begins only with the polynomial-specific exponent census: cyclic run avoidance over irreducible orbits and the resulting all-depth and target-fibre formulas.

Shifted gcds and shift classes also occur in symbolic summation and shiftless factorization [2]; those algorithmic interfaces, like standard transfer matrices, are background rather than novelty claims.

2. TRANSLATION ORBITS AND LOCAL DEPTH

Translation acts on monic irreducibles in orbits of length 1 or p . Fix a nonfixed orbit and choose an indexing

$$P_0, P_1 = \sigma P_0, \dots, P_{p-1} = \sigma^{p-1} P_0,$$

with indices read modulo p . If e_i is the exponent of P_i in f , then, up to reversing the cyclic indexing, (5) says that its exponent after t steps is

$$(6) \quad e_i^{(t)} = \min_{0 \leq j \leq t} e_{i+j}.$$

At $t = p-1$ every coordinate equals $m = \min_i e_i$. Put $c_i = e_i - m$. Then $c_i \geq 0$ and $\min_i c_i = 0$.

For a cyclic vector $c = (c_0, \dots, c_{p-1})$ with a zero, let $\lambda(c)$ denote the longest cyclic run of positive coordinates; set $\lambda(0, \dots, 0) = 0$.

Lemma 2.1 (Local run statistic). *On a nonfixed irreducible orbit, the least t for which the exponents in (6) have all reached their terminal value m is $\lambda(c)$.*

Proof. After subtracting m , equation (6) vanishes in every coordinate exactly when every cyclic interval of length $t+1$ contains a zero. This is equivalent to the absence of a positive cyclic run of length $t+1$, or to $\lambda(c) \leq t$. The least such t is $\lambda(c)$, including the all-zero boundary. $\square$

Fixed irreducibles contribute only to the invariant core and have no local transient. Hence the global depth of f is the maximum of $\lambda(c)$ over its nonfixed irreducible orbits, with the maximum of an empty family equal to zero.

3. THE ALL-DEPTH EULER PRODUCT

For $0 \leq t \leq p-1$, define the local residual series

$$(7) \quad R_{p,t}(y) = \sum_{\substack{c \in \mathbb{N}^p, \min_i c_i = 0 \\ \lambda(c) \leq t}} y^{c_0 + \dots + c_{p-1}}.$$

It has a finite transfer description. Put $u = y/(1-y)$ and index rows and columns by $0, \dots, t$. Let $M_t(u)$ be the $(t+1) \times (t+1)$ matrix whose only nonzero entries are

$$(8) \quad (M_t)_{i,0} = 1 \quad (0 \leq i \leq t), \quad (M_t)_{i,i+1} = u \quad (0 \leq i < t).$$

Lemma 3.1 (Cyclic transfer). *For $0 \leq t \leq p-1$,*

$$(9) \quad R_{p,t}(y) = \text{tr}(M_t(u)^p).$$

Proof. Read a cyclic support word of length p . State i records the current positive run length. A zero sends every state to 0 with weight 1; a positive coordinate sends i to $i+1$ with weight $u = y + y^2 + \dots$, provided $i < t$. A closed length- p path is therefore a cyclic word with no positive run longer than t , together with an arbitrary positive height at every positive position.

Every admissible support has a zero. Starting just after any zero and propagating run lengths gives a unique state at every cyclic cut, hence a unique closed state path; the all-zero support likewise has the unique all-zero path. Conversely every closed path contains a zero transition and recovers this support and its heights. Summing closed paths is the trace in (9), and their weights are exactly those in (7). $\square$

Write $[z^n]F(z)$ for coefficient extraction in formal power series.

Theorem 3.2 (All-depth Euler product). *For every prime power $q = p^a$ and $0 \leq t \leq p-1$, the ordinary generating function for monic polynomials of depth at most t is*

$$(10) \quad H_{q,p,t}(z) := \sum_{f \in \mathcal{M}_q: \text{depth}(f) \leq t} z^{\deg f} = \frac{1}{1 - qz^p} \prod_{d \geq 1} R_{p,t}(z^d)^{a_d},$$

where a_d is the owned input in (4) and $R_{p,t}$ is given by (9). The product is coefficientwise well defined.

Proof. Unique factorization partitions the irreducible factors of f into fixed orbits and length- p orbits. On each length- p orbit, the exponent vector splits uniquely as

$$(e_0, \dots, e_{p-1}) = m(1, \dots, 1) + c, \quad \min_i c_i = 0.$$

All common-minimum pieces, together with every fixed irreducible factor, form one arbitrary translation-invariant monic polynomial. Their series is $I_{q,p}(z)$ from (1). For each of the a_d nonfixed degree- d orbits, lemma 2.1 says that depth at most t is equivalent to $\lambda(c) \leq t$, and the residual degree is $d \sum_i c_i$. Its series is therefore $R_{p,t}(z^d)$.

The global depth is the maximum of the local depths, so these conditions hold independently on every orbit. Multiplication yields (10). For a fixed coefficient z^n , only irreducible degrees $d \leq n$ can contribute, proving formal well-definedness. $\square$

Corollary 3.3 (Exact depth layers). *Set $H_{q,p,-1}(z) = 0$. For $0 \leq t \leq p-1$,*

$$\#\{f \in \mathcal{M}_q : \deg f = n, \text{depth}(f) = t\} = [z^n](H_{q,p,t}(z) - H_{q,p,t-1}(z)).$$

Moreover,

$$(11) \quad R_{p,0}(y) = 1, \quad H_{q,p,0}(z) = \frac{1}{1 - qz^p}, \quad R_{p,p-1}(y) = \frac{1 - y^p}{(1 - y)^p}, \quad H_{q,p,p-1}(z) = \frac{1}{1 - qz}.$$

Proof. The difference isolates the two nested depth thresholds. At $t = 0$, the only normalized vector is zero. At $t = p-1$, every nonnegative vector with minimum zero is allowed, so its series is the series of all vectors minus that of all-positive vectors:

$$(1 - y)^{-p} - (y/(1 - y))^p = (1 - y^p)/(1 - y)^p.$$

The first global identity in (11) is (1); the last follows either from theorem 3.2 and unique factorization or simply because every monic polynomial has depth at most $p-1$ and there are q^n of degree n . $\square$

4. UNIT FIBRE AND EVERY INVARIANT TARGET

The endpoint $Q(f)$ divides f and is invariant. Define the *unit fibre*

$$\mathcal{U}_{q,p} = Q^{-1}(1)$$

and let $U_{q,p,n}$ count its degree- n elements. The word “unit” refers only to the target 1.

Proposition 4.1 (Graded terminal split). *Multiplication gives a degree-preserving bijection*

$$(12) \quad \{h \in \mathcal{M}_q : \sigma h = h\} \times \mathcal{U}_{q,p} \longrightarrow \mathcal{M}_q, \quad (h, r) \longmapsto hr.$$

Its inverse is $f \mapsto (Q(f), f/Q(f))$. This is a set-theoretic graded bijection, not a monoid quotient.

Proof. On every nonfixed irreducible orbit, $Q(f)$ has the common minimum exponent; on a fixed irreducible it has the full exponent. Hence $h = Q(f)$ is invariant, divides f , and the residual $r = f/h$ has minimum exponent zero on every nonfixed orbit and no fixed irreducible factor. Thus $Q(r) = 1$. This proves existence and also forces both factors, proving uniqueness.

Equivalently, for invariant h equation (5) gives the restricted identity

$$(13) \quad Q(hr) = hQ(r).$$

This identity proves that every pair on the left of (12) maps back to itself. It does not imply multiplicativity on arbitrary pairs. $\square$

Theorem 4.2 (Unit fibre and target fibres). *The unit-fibre generating function and coefficients are*

$$(14) \quad U_{q,p}(z) := \sum_{n \geq 0} U_{q,p,n} z^n = \frac{1 - qz^p}{1 - qz},$$

$$(15) \quad U_{q,p,n} = \begin{cases} q^n, & 0 \leq n < p, \\ q^n - q^{n-p+1}, & n \geq p. \end{cases}$$

Let h be any invariant monic polynomial of degree m . Then, for every $N \geq 0$,

$$(16) \quad \#\{f \in \mathcal{M}_q : \deg f = N, Q(f) = h\} = \begin{cases} U_{q,p,N-m}, & N \geq m, \\ 0, & N < m. \end{cases}$$

Writing $L = D - m$, the degree-capped fibre through D is zero for $L < 0$; for $L \geq 0$ it equals

$$(17) \quad \sum_{n=0}^L U_{q,p,n} = \begin{cases} \frac{q^{L+1} - 1}{q - 1}, & 0 \leq L < p, \\ \frac{q^{L+1} - 1 - q^{L-p+2} + q}{q - 1}, & L \geq p. \end{cases}$$

Thus fibres depend on the target only through its degree.

Proof. All monic polynomials have series $(1 - qz)^{-1}$. Taking generating functions in the graded bijection (12) and using (1) gives

$$\frac{1}{1 - qz} = \frac{1}{1 - qz^p} U_{q,p}(z),$$

which is (14); coefficient extraction gives (15).

For a fixed invariant h , multiplication $r \mapsto hr$ maps the degree- $(N - m)$ part of $\mathcal{U}_{q,p}$ into the fibre over h by (13). Conversely, proposition 4.1 forces every element of that fibre to be hr for a unique such r . This proves (16). Summing (15) through L gives (17); the second case subtracts $q \sum_{j=0}^{L-p} q^j$ from $\sum_{j=0}^L q^j$. $\square$

Remark 4.3 (Why this is not a kernel). The projection Q is not a monoid homomorphism. In characteristic p , put $a = x$ and $b = (x^p - x)/x$. The two exponent supports each miss a member of the full translation orbit, so $Q(a) = Q(b) = 1$, whereas $ab = x^p - x$ is invariant and $Q(ab) = x^p - x$. Hence $Q^{-1}(1)$ is a unit fibre, not an algebraic kernel. The proofs above use only orbit exponents and the restricted identity (13).

5. MECHANICAL FALSIFICATION AND LIMITATIONS

An independent verifier constructs

$$\mathbb{F}_4 = \mathbb{F}_2[u]/(u^2 + u + 1), \quad \mathbb{F}_8 = \mathbb{F}_2[u]/(u^3 + u + 1), \quad \mathbb{F}_9 = \mathbb{F}_3[u]/(u^2 + 1)$$

from explicit field tables. It implements literal Horner translation, Euclidean gcd, trial-division irreducibility, translation-orbit partitioning, direct residual-vector enumeration, target fibres, and the Euler coefficients without importing the historical prime-field engine. It checks all monic polynomials through degrees 6, 4, 4, respectively: 17,523 states and 180,403 deterministic assertions. In particular, the $\mathbb{F}_9$ lane exercises the intermediate depths 0, 1, 2. This finite enumeration is evidence against small counterexamples, never a proof of theorems 3.2 and 4.2.

The result is limited to the single order- p translation generated by 1. Characteristic zero has no such finite clock; simultaneous erosion by a larger additive subspace is a different map. We make no priority claim for the literal dynamics, (5), the invariant ring, the fixed irreducible formula, or the transfer method. The owner-subtracted residual is precisely the conjunction of theorem 3.2 with the target-refined theorem 4.2. A bounded literature non-hit is not novelty clearance, and external posting or submission remains on hold pending specialist owner review and explicit authorization.

6. CONCLUSION

After the known orbit fold and fixed-irreducible theory are removed, the transient census is controlled by one local statistic: the longest positive run in a normalized cyclic exponent vector. Its transfer series produces all depth thresholds at once, while the same normalization gives a uniform graded fibre above every invariant endpoint. These two formulas constitute the complete internal claim package; the older clock, fixed set, and owner formulas remain background.

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